

OLPROC

An Observation List Processor
for use with
PEST and PEST++

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1. Introduction

1.1 General

OLPROC stands for “observation list processor”. Its name complements those of other members of a suite of programs that facilitates use PEST and PEST++ in conjunction with groundwater models in general, and the MODFLOW family of models in particular. These other programs are named PLPROC and TS6PROC; they are documented in their own manuals.

In contrast to PLPROC and TS6PROC, OLPROC’s attention is focussed on observations rather than on parameters. OLPROC can run in two modes. In one of these modes it builds part of a PEST input dataset, including part of a PEST control file together with instruction files that are cited in the PEST control file. In its other mode, it runs as a model postprocessor invoked through a batch or script file that PEST runs as “the model”. When run in this capacity it reads model output files; OLPROC then writes another set of files containing numbers which are directly comparable with field observations.

1.2 Running OLPROC

Regardless of its operating mode, OLPROC reads an “OLPROC control file”, the details of which are explained in this manual; it also reads a number of user-prepared and model-generated files whose names are cited in its control file. When run in “postprocessor” mode, OLPROC writes a small number of files, all of which are meant to be read by PEST. When run in “construction” mode, it also writes the instruction files which allow PEST to read these files, together with a partial PEST control file containing the *observation groups*, *observation data* and *model input/output* sections of a complete PEST control file.

To run OLPROC in “construction” mode, type the following command at the screen prompt.

```
olproc filename
```

In the above command, *filename* is the name of an OLPROC control file. Alternatively, type the command:

```
olproc filename 0
```

or

```
olproc filename /0
```

In the above commands, “0” (i.e. zero) signifies OLPROC’s mode of operation.

To run OLPROC as a model postprocessor, type either of the following commands at the screen prompt, or include one of these commands in a batch or script file.

```
olproc filename 1
```

or

```
olproc filename /1
```

As an alternative to the above commands, you may just type:

```
olproc
```

at the screen prompt. OLPROC will then ask:

```
Enter name of OLPROC control file:  
Enter OLPROC mode index [0 or 1]:
```

If the command to run OLPROC is issued from a batch or script file, and if the name of its control file and the index of its mode of operation are not provided on its command line (as exemplified in the second of the above options), then these can be provided in a text file. Suppose that this file is named *olproc.in*. OLPROC can be directed to accept input from this file rather than from the keyboard using the command:

```
OLPROC < olproc.in
```

2. Files Containing Data

2.1 General

When run in either of its modes, OLPROC matches model outputs to field measurements. The file types through which both of these data types must be supplied are described in this chapter.

As is described below, field measurements comprising a history-matching dataset must be accompanied by dates, and optionally by times. Two formats are supported for dates. These are:

`dd/mm/yyyy`

and

`mm/dd/yyyy`

Examples of these protocols are respectively:

`25/01/2021`

and

`01/25/2021`

If desired, leading zeros can be omitted, so that the above date representations become:

`25/1/2021`

and

`1/25/2021`

It is incumbent on an OLPROC user to inform OLPROC which of the above date protocols he/she intends to employ. This is done through an entry in its control file. The former protocol is adopted in the present document. However, in everyday OLPROC usage, either protocol can be adopted. Whichever protocol is selected, consistency in its use is essential.

Optionally, a measurement can be accompanied by a time. This must be supplied using the protocol:

`hh:mm:ss`

Examples are:

`00:00:00`

`12:05:42`

Leading zeroes can be omitted if desired. The above times then become:

`0:0:0`

`12:5:42`

2.2 Measurements

2.2.1 General

A user can supply field measurements to OLPROC using either or both of two different file types. These are now described.

2.2.2 CSV Files

As often as not, modellers, and/or those who assist modellers to manage data, use a spreadsheet program such as Microsoft EXCEL to store field measurements. EXCEL allows each sheet of stored data to be saved as a CSV file. Provided that certain protocols are respected, OLPROC can read this CSV file.

Figure 2.1 shows the top left portion of a single sheet of data stored in Microsoft EXCEL. If this sheet is saved as a CSV file, it can be read by OLPROC.

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Layer	1		1		1		2		1		3	
2	Target Name	212392K01		212632A01		222610J01		212501F01		222423J01		222525N01	
3	Group Name	GWH		GWH		GWH		GWH		GWH		GWH	
4	Easting and	6412096	1903994	6508784	1906873	6519177	1893213	6495646	1931700	6461034	1883355	6494108	1875965
5	Use absolutes	Yes		Yes		Yes		Yes		Yes		Yes	
6	Use differences	Yes		Yes		Yes		Yes		Yes		Yes	
7	Type	RMS and Calibration		RMS and Calibration		Calibration		Calibration		RMS and Calibration		RMS and Calibration	
8	GSA	LTRID		LTRID		LTRID		LTRID		Pixley		Pixley	
9	Page	2		6		6		5		3		10	
10		Date	GW_Elev	Date	GW_Elev	Date	GW_Elev	Date	GW_Elev	Date	GW_Elev	Date	GW_Elev
11		8/10/1986	199.1	1/10/1986	299.0	2/10/1986	300.8	29/01/1987	293.1	1/10/1986	125.0	23/01/1986	161.0
12		1/01/1987	198.9	10/10/1986	303.5	30/01/1987	303.6	5/10/1987	273.6	1/01/1987	129.0	28/09/1987	135.0
13		5/02/1987	198.4	27/01/1987	306.9	1/04/1987	297.7	28/01/1988	280.5	16/01/1987	141.0	1/02/1988	167.0
14		1/04/1987	190.8	1/10/1987	285.6	1/07/1987	288.0	22/09/1988	261.7	1/04/1987	129.0	1/04/1988	106.0
15		1/07/1987	180.9	6/10/1987	290.1	6/10/1987	284.4	24/01/1989	268.6	1/07/1987	122.5	1/07/1988	116.0
16		9/10/1987	188.3	28/01/1988	294.9	28/01/1988	295.7	3/10/1989	248.9	28/09/1987	120.0	30/09/1988	130.5
17		4/02/1988	192.5	29/09/1988	280.2	1/04/1988	287.9	13/02/1990	254.2	1/01/1988	120.0	25/01/1989	122.0
18		1/04/1988	182.0	1/10/1988	275.7	1/07/1988	276.7	4/10/1990	229.4	29/01/1988	126.0	1/04/1989	127.0
19		1/07/1988	156.1	23/01/1989	282.9	28/09/1988	275.8	30/01/1991	230.4	1/04/1988	121.0	1/07/1989	104.0
20		3/10/1988	175.7	1/10/1989	268.6	25/01/1989	280.8	7/10/1991	216.8	1/07/1988	109.0	2/10/1989	131.0
21		27/01/1989	183.0	3/10/1989	273.1	4/10/1989	267.1	29/01/1992	228.9	30/09/1988	114.0	30/01/1990	142.5
22		1/04/1989	169.0	14/02/1990	274.1	14/02/1990	273.3	14/10/1992	205.9	1/01/1989	116.0	1/04/1990	115.0
23		9/10/1989	170.9	1/10/1990	255.8	1/07/1990	251.4	26/01/1993	218.5	25/01/1989	117.5	1/07/1990	78.0
24		1/01/1990	170.8	2/10/1990	260.3	4/02/1991	259.2	7/10/1993	232.3	1/04/1989	112.5	5/10/1990	108.0
25		26/02/1990	153.5	4/02/1991	259.5	1/04/1991	253.7	26/01/1994	238.7	1/07/1989	103.5	28/01/1991	125.0
26		1/04/1990	153.1	30/09/1991	244.2	4/10/1991	245.4	3/10/1994	211.0	1/10/1989	98.5	1/04/1991	115.0
27		1/07/1990	145.0	1/10/1991	239.7	28/01/1992	253.5	31/01/1995	225.4	4/10/1989	100.5	1/07/1991	43.0
28		5/10/1990	153.3	27/01/1992	254.5	1/04/1992	244.8	3/10/1995	255.3	1/01/1990	106.0	30/09/1991	63.0
29		1/01/1991	154.8	1/10/1992	227.0	20/10/1992	241.7	1/02/1996	264.1	26/01/1990	110.0	4/02/1992	114.0
30		1/04/1991	135.5	13/10/1992	231.5	17/02/1993	245.2	2/10/1996	271.0	1/04/1990	99.0	1/04/1992	109.0
31		7/10/1991	137.7	8/02/1993	236.5	1/04/1993	249.4	29/01/1997	274.3	1/07/1990	90.0	1/07/1992	97.0
32		1/01/1992	138.5	1/10/1993	229.6	1/07/1993	250.5	15/10/1997	263.0	1/10/1990	90.0	5/10/1992	72.0
33		1/04/1992	121.0	6/10/1993	234.1	18/10/1993	255.2	27/01/1998	270.6	2/10/1990	92.0	27/01/1993	103.0

Figure 2.1. The top left of a sheet of data stored in Microsoft EXCEL.

As is apparent from Figure 2.1, the sheet is partitioned into two sections. The top section of the sheet comprises headers, while the bottom section of the sheet records measurement data. In the above example, two columns are devoted to the recording of measurements made at each of a number of sites. The first of these columns contains measurement dates, while the second of these columns contains the measurements themselves. As will be discussed below, an optional third column can contain measurement times.

The following specifications are critical.

1. The beginning of the measurement component of the sheet is marked by the text string “date” in the sheet’s second column. In the above example, this occurs in row 10. Where measurement times are not supplied with measurement dates, then other text in other columns of this row are ignored by OLPROC. However OLPROC assumes that every second column below this row contains dates, and that the column to its immediate right contains measurements associated with those dates.
2. The headers listed in the first section of the sheet in rows that precede those in which measurements are stored must be vertically aligned with the measurements to which they pertain. Optionally, neighbouring rows can be amalgamated in this section; if they are not amalgamated, the text header string must be recorded in the first of the columns

that are associated with measurements pertaining to a particular site - that is, the column containing measurement dates.

3. The first portion of the sheet can include as many header rows as a user desires. OLPROC ignores all of these rows except for the following:

target name

group name

use absolutes

use differences

The above text strings must be recorded in the first column of these rows. They can be supplied in arbitrary order. Note also that OLPROC is not sensitive to case. Information that is recorded in the rows that commence with these strings must follow in every second column, or in amalgamated cells aligned with ensuing data columns.

4. A “target” is a measurement site. In accordance with PEST protocol, its name must be 20 characters or less in length.
5. PEST requires that each observation be assigned to a group. In the PEST dataset that is constructed by OLPROC, all measurements that were made at a particular site are assigned to the group with which the site is associated. Hence group names recorded in the first section of the sheet are transferred directly to the PEST input dataset. Even though PEST allows observation group names to have up to 12 characters, OLPROC insists that group names be restricted to 10 characters in length. The reason for this will become apparent soon.
6. In the data section of the sheet (i.e. in the second section), the dates associated with measurements must increase with row number. Apart from this requirement, dates are arbitrary. They do not need to be the same for different sites (i.e. different targets).
7. The number of measurements associated with one particular site can be different from that associated with other sites.
8. The name of a site must not be confusable with a number. The reason for this will also become clear later in this manual. Hence OLPROC will declare the following target names to be illegal:
 - 12345
 - 1234E23
 - 1234D54
 - 1234Q54

OLPROC will cease execution with an appropriate error message if any of the above specifications are violated.

Figure 2.2 shows an excerpt from another spreadsheet page. This sheet includes measurement times. The protocols for reading the CSV file that is saved from this sheet are very similar to those described above. The beginning of the measurement section of the spreadsheet is still marked by the string “date” in the second column. However the string “time” must occupy the cell immediately to the right of that containing “date”. All other contents of this row are ignored. The presence of “time” immediately following “date” informs OLPROC that data columns are grouped into sets of three, with dates recorded in the first of each set of three columns, times in the second, and field measurements in the third. For each measurement site,

dates and times must increase down the page. The number of measurements associated with different sites can be different. The set of measurement dates and times that correspond to one site can differ from that corresponding to another site.

	A	B	C	D	E	F	G	H	I	J
1	Layer	1			1			1		
2	Target Name	212332K01			212632A01			222610J01		
3	Group Name	GWH			GWH			GWH		
4	Easting and Northing	6412096	1903994		6508784	1906873		6519177	1893213	
5	Use absolutes	Yes			Yes			Yes		
6	Use differences	Yes			Yes			Yes		
7	Type	RMS and Calibration			RMS and Calibration			Calibration		
8	GSA	LTRID			LTRID			LTRID		
9	Page	2			6			6		
10		Date	Time	GW_Elev	Date	Time	GW_Elev	Date	Time	GW_Elev
11		8/10/1986	11:34:0	199.1	1/10/1986	15:00:00	299.0	2/10/1986	8:25:21	300.8
12		1/01/1987	12:00:00	198.9	10/10/1986	15:23:00	303.5	30/01/1987	14:23:12	303.6
13		5/02/1987	9:23:00	198.4	27/01/1987	12:04:00	306.9	1/04/1987	15:23:21	297.7
14		1/04/1987	8:30:00	190.8	1/10/1987	10:40:00	285.6	1/07/1987	8:23:00	288.0
15		1/07/1987	13:00:00	180.9	6/10/1987	11:28:00	290.1	6/10/1987	14:23:00	284.4
16		9/10/1987	15:23:00	188.3	28/01/1988	12:50:00	294.9	28/01/1988	11:34:0	295.7
17		4/02/1988	17:23:00	192.5	29/09/1988	19:32:00	280.2	1/04/1988	12:00:00	287.9
18		1/04/1988	11:00:00	182.0	1/10/1988	12:23:00	275.7	1/07/1988	9:23:00	276.7
19		1/07/1988	8:00:00	156.1	23/01/1989	8:25:21	282.9	28/09/1988	8:25:21	275.8
20		3/10/1988	14:40:00	175.7	1/10/1989	14:23:12	268.6	25/01/1989	14:23:12	280.8
21		27/01/1989	19:25:21	183.0	3/10/1989	15:23:21	273.1	4/10/1989	15:23:21	267.1
22		1/04/1989	14:00:00	169.0	14/02/1990	8:23:00	274.1	14/02/1990	8:23:00	273.3
23		9/10/1989	15:00:00	170.9	1/10/1990	14:23:00	255.8	1/07/1990	14:23:00	251.4
24		1/01/1990	15:23:00	170.8	2/10/1990	11:34:0	260.3	4/02/1991	11:34:0	259.2
25		26/02/1990	12:04:00	153.5	4/02/1991	12:00:00	259.5	1/04/1991	12:00:00	253.7
26		1/04/1990	10:40:00	153.1	30/09/1991	9:23:00	244.2	4/10/1991	9:23:00	245.4
27		1/07/1990	11:28:00	145.0	1/10/1991	8:30:00	239.7	28/01/1992	8:30:00	253.5
28		5/10/1990	12:50:00	153.3	27/01/1992	13:00:00	254.5	1/04/1992	13:00:00	244.8
29		1/01/1991	19:32:00	154.8	1/10/1992	15:23:00	227.0	20/10/1992	15:23:00	241.7
30		1/04/1991	12:23:00	135.5	13/10/1992	17:23:00	231.5	17/02/1993	14:23:12	245.2
31		7/10/1991	8:25:21	137.7	8/02/1993	11:00:00	236.5	1/04/1993	15:23:21	249.4

Figure 2.2. A spreadsheet in which times accompany dates.

Figure 2.3 shows the top left of the CSV file that is recorded when the sheet appearing in Figure 2.1 is saved in this format.

Figure 2.3. The top left of the sheet appearing in Figure 2.1 when saved as a CSV file.

OLPROC allows a user to construct a PEST and/or PEST++ input dataset in which:

- Models are often better at calculating differences between system states and fluxes than the direct values of system states and fluxes. The latter are referred to as “absolutes” in the present discussion. Model-calculated differences often suffer less from biases induced by model simplifications and defects. Hence the matching of differences between field measurements on the one hand with model-calculated counterparts to these differences on the other hand, often reduces the proclivity for parameters to adopt surrogate roles which compensate for model defects as they are constrained by the history-matching process.

The first section of the spreadsheet page from which an OLPROC-ready CSV file is recorded must include two rows which are labelled as *use differences* and *use absolutes* (or *use_differences* and *use_absolutes*). A value of “yes” or “no” must be ascribed to every measurement site. If, for a particular measurement site, the value of *use differences* is “yes”, then OLPROC calculates the difference between every measurement and the first measurement

at that site that lies within the history-matching time window. (See section 4 for the definition of “history-matching time window”.) These “difference measurements” are then included in the partial PEST control file which OLPROC writes. When run in both of “construction” and “postprocessor” modes, OLPROC calculates differences between corresponding model-calculated quantities after interpolation to the same times at which measurements were made.

2.2.4 Site Names and Group Names

The first section of a measurement CSV file must also include rows labelled *target name* and *group name* (or *target_name* and *group_name*). The first of these awards a name to each measurement site. The second provides the name of an observation group to which measurements from that site will be ascribed in the PEST input dataset that is written by OLPROC.

As already stated, PEST observation names must be 20 characters or less in length. PEST forms the name of each non-difference (i.e. “absolute”) observation by appending the suffix “_N” to the site (i.e. target) name, where *N* is its sequence number, starting at 1. Before forming these observation names, however, it truncates the site name to a length that accommodates representation of the highest number in the sequence. If site names are similar, this can cause a coincidence of observation names. OLPROC will report such a coincidence if it occurs. However users are advised to avoid this occurrence by restricting site names to a length which is short enough to avoid loss of trailing characters through truncation in this manner.

Where measurement differences are used as PEST observations, each observation name is formed by adding the suffix “_dN” to the name of the pertinent site. Numbering begins at 2, as the first difference is taken between the second user-supplied measurement within the history-matching time window and the first user-supplied measurement within this window. Obviously, inclusion of “d” within the observation name suffix requires one extra character. The user should take into account the potential for observation name ambiguity which this may engender when naming measurement sites.

In contrast to observations, PEST limits the names of observation groups to 12 characters or less in length. As has already been discussed, OLPROC limits the names of observation groups to 10 characters or less in length. Absolute observations pertaining to a particular measurement site are assigned to the user-supplied observation group associated with that site. Differenced observations are assigned to an observation group whose name is formed from the user-supplied observation group by appending the suffix “_d” to the name of that group.

2.2.5 Other CSV File Headers

As is apparent from Figures 2.1 and 2.2, the first section of a user-supplied measurement CSV file may include many header rows. OLPROC ignores these. It reads only the rows discussed in the previous two sections, that is those that are labelled *use absolutes*, *use differences*, *target name* and *group name*. It requires that all of these rows be present.

2.3 Site Sample Files

Site sample files are denoted as “SSF files” in this document. They are described in documentation of the PEST Groundwater Utilities Suite. However, as is described below, OLPROC’s expectations of these files are a little less restrictive than those of older versions of the Groundwater Utilities.

Site sample files are space or tab-delimited text files. Figure 2.4 shows an example.

#	site	date	time	head
	212332k01	01/10/1986	12:00:00	197.50194
	212332k01	01/11/1986	12:00:00	192.12872
	212332k01	01/12/1986	12:00:00	189.63802
	212332k01	01/01/1987	12:00:00	188.19571
	212332k01	01/02/1987	12:00:00	187.24901
	212332k01	01/03/1987	12:00:00	186.26927
	212332k01	01/04/1987	12:00:00	185.60912
	212332k01	01/05/1987	12:00:00	184.35678
	212332k01	01/06/1987	12:00:00	181.84254
	...			
	222525n01	01/12/1989	12:00:00	91.135010
	222525n01	01/01/1990	12:00:00	151.81700
	222525n01	01/02/1990	12:00:00	171.36487
	222525n01	01/03/1990	12:00:00	178.20529
	222525n01	01/04/1990	12:00:00	181.80038
	222525n01	01/05/1990	12:00:00	182.82175
	222525n01	01/06/1990	12:00:00	107.81244
	222525n01	01/07/1990	12:00:00	55.911480
	222525n01	01/08/1990	12:00:00	50.641434
	...			
	222420a01	01/06/1999	12:00:00	73.839310
	222420a01	01/07/1999	12:00:00	69.870567
	222420a01	01/08/1999	12:00:00	66.045868
	222420a01	01/09/1999	12:00:00	62.616489
	222420a01	01/10/1999	12:00:00	59.073528
	222420a01	01/11/1999	12:00:00	56.887295
	222420a01	01/12/1999	12:00:00	56.681877
	222420a01	01/01/2000	12:00:00	56.952751
	222420a01	01/02/2000	12:00:00	58.681473
	etc			

Figure 2.4 Example of a site sample file.

As Figure 2.4 shows, a site sample file contains four columns of data. The first column contains the names of measurement sites (i.e. targets). The second and third columns contain dates and times respectively. The fourth column contains measurements.

The following rules apply.

1. Measurements pertaining to any particular site must be contiguous. That is, all data pertaining to site A must precede all data pertaining to site B, etc.
2. For any site, measurements must be supplied in increasing order of time.
3. The name of a site can be 20 characters or less in length. (However fewer characters than this are recommended because of the way in which OLPROC forms PEST observation names from site names.)
4. There is no limit to the number of sites that can feature in a single site sample file.
5. Comment lines can appear anywhere within an SSF file. Comment lines are recognized by the fact that their first non-blank character is "#". To avoid confusion, a site name should therefore not begin with "#".
6. Blank lines can appear anywhere within a site sample file.

OLPROC can read both measured and model-generated data from SSF files. A user can prepare a measurement SSF file using a text editor, a spreadsheet or a self-written program. Utility

programs supplied with the PEST Groundwater Utilities can postprocess model outputs to produce site sample files containing model-calculated states and fluxes. They can also undertake useful simulation support tasks such as spatial and temporal interpolation of data contained in these files. See, for example, MOD2OBS, USGMOD2OBS, SMP2SMP, PESTPREP1, PESTPREP2 and other programs from this suite.

2.4 MODFLOW 6 OBS Output Files

All packages of the MODFLOW 6 simulator support observation (OBS) functionality. In the input file for a particular package, a user denotes sites and data types which he/she would like recorded in an output file. Depending on the data type, the site can incorporate model-calculated quantities pertaining to one or many cells. Hence, for example, time-varying heads at a single cell centre, or time-varying fluxes through a group of cell-based boundary conditions, can be demarcated for recording. These model-calculated quantities are recorded at the end of each model time step in either a CSV or binary file. Quantities calculated for many sites can be recorded in a single file. Many such files can be recorded.

As presently programmed, OLPROC reads MODFLOW 6 OBS-generated CSV files; it does not read OBS-generated binary files. The first part of a MODFLOW 6 OBS-generated CSV file is shown in Figure 2.5.

```
time,ID1_5_2_GWHEAD,ID2_5_2_GWHEAD
2628000.000000,1018.180726,1018.047126
2803200.000000,1018.045900,1018.045900
2978400.000000,1018.044678,1018.044678
3153600.000000,1018.043456,1018.043456
3328800.000000,1018.042232,1018.042232
3504000.000000,1018.041008,1018.041008
3679200.000000,1018.039783,1018.039783
3854400.000000,1018.038560,1018.038560
4029600.000000,1018.037340,1018.037340
4204800.000000,1018.036123,1018.036123
4380000.000000,1018.034910,1018.034910
4555200.000000,1018.033703,1018.033703
etc
```

Figure 2.5. First part of a MODFLOW 6 OBS output file.

MODFLOW 6 allows a user to choose the number of significant figures with which numbers are represented in an OBS-generated CSV file. It is wise to request a high number of significant figures (9 or 10) so that finite differences formed by PEST to calculate parameter sensitivities have integrity.

MODFLOW 6 requires that a user associate names with observation sites. If OLPROC must read these files, ensure that these names are 20 characters or less in length.

3. The OLPROC Control File

The present and following sections of this manual describe the OLPROC control file. This file can easily be prepared by an OLPROC user with the help of his/her favourite text editor. In describing the contents of this file, OLPROC functionality is also described.

An OLPROC control file is divided into blocks. Each block begins with:

```
BEGIN blockname
```

or

```
START blockname
```

where *blockname* is the name of the block. The end of a block is identified by the text:

```
END blockname
```

Note that all components of an OLPROC input file are case-insensitive, with the exception of filenames if working on a LINUX system.

The names of blocks that can appear in an OLPROC control file are listed in Table 3.1. The GENERAL block must be supplied first. Other blocks (if present) can be supplied in any order.

Block name	Contents
GENERAL	Information required to process contents of all other blocks.
MEAS_CSV_FILES	The names of CSV files containing field measurements.
MEAS_SSF_FILES	The names of SSF files containing field measurements.
MF6_OBS_FILES	The names of MODFLOW 6 OBS files containing model outputs.
MODEL_SSF_FILES	The names of SSF files containing model outputs.
SECONDARY_MODEL_OUTCOMES	Equations through which secondary model outcomes are calculated from primary model outcomes.
OUTPUT	Some details affecting OLPROC output files.
WEIGHTS	Information on how weights recorded in an OLPROC-generated partial PEST control file are formulated.

Table 3.1. Blocks within an OLPROC control file.

Most OLPROC blocks are optional. Only the GENERAL and OUTPUT blocks are essential. However an OLPROC control file must contain either a MEAS_CSV_FILES block or a MEAS_SSF_FILES block (or both). It must also contain either a MODEL_SSF_FILES block or a MF6_OBS_FILES block (or both).

As will be demonstrated below, in most blocks an item of OLPROC input data must be supplied following a keyword which identifies that item. Optionally, a “=” symbol can separate the keyword from the data item. Within any block, some keywords are optional while some are

mandatory. OLPROC will cease execution with an appropriate error message if a mandatory keyword is omitted.

Comment lines and blank lines can be placed anywhere within an OLPROC control file. A comment line begins with the “#” character. Note that the “#” character must appear at the beginning of a line. If it appears elsewhere on the line, then it is interpreted as a normal character. Note also that, to avoid confusion in reading any of its input files, OLPROC does not allow a site name to begin with the “#” character.

Figure 3.1 shows an example of an OLPROC control file.

```

# An OLPROC control file.

START GENERAL
  date_format      = dd/mm/yyyy
  time_for_csv_files = 12:00:00
  model_start_date  = 1/1/1986
  model_time_units  = days
END GENERAL

START MEAS_CSV_FILES
  file = head.csv
  file = flux.csv
END MEAS_CSV_FILES

START MEAS_SSF_FILES
  file = gwh1.ssf  group=layer1 use_abs=yes use_diff=yes
  file = gwh2.ssf  group=layer2 use_abs=yes use_diff=yes
  file = gwf.ssf   group=flux   use_abs=yes use_diff=no
END MEAS_SSF_FILES

START MF6_OBS_FILES
  file = sfr.reach29.csv
  file = obs_lak.csv
END MF6_OBS_FILES

START MODEL_SSF_FILES
  file = gwh_model.smp
  file = sub_model.smp
END MODEL_SSF_FILES

START SECONDARY_MODEL_OUTCOMES
  ff1= flux1 + flux2 + flux3
  hd1 = 0.5 * (bh3213 + bh3982)
  hd2 = min(head5,0.0)
  depth = 33.213 - (head1 + head2)/2.0
END SECONDARY_MODEL_OUTCOMES

START OUTPUT
  partial_pest_control_file = partial.pst
  model_batch_file = runmodel.bat
  obs_ssf_folder = .\obs_files
END OUTPUT

START WEIGHTS
  name          = drawdown          &
  date1         = 1/1/1990 date2=1/1/2000 &
  weight        = 1.0/(obsval+0.005) &
  minweight     = 0.001              &
  maxweight     = 10.0               &
  type          = diff
  name = heads type=diff weight=5.0
END WEIGHTS

```

Figure 3.1 Example of an OLPROC control file.

4. GENERAL Block

4.1 General

The GENERAL block must be the first block appearing in an OLPROC control file. Keywords which can be used in the GENERAL block are listed in Table 4.2.

Keyword	Status	Values
<i>date_format</i>	mandatory	<i>dd/mm/yyyy</i> or <i>mm/dd/yyyy</i>
<i>time_for_csv_files</i>	optional	A time of day; default is <i>12:00:00</i>
<i>model_time_units</i>	optional	<i>days</i> or <i>seconds</i> ; default is <i>days</i>
<i>time_base_file</i>	optional	The name of a file; the default is no file
<i>model_start_date</i>	mandatory	A date in format dictated by <i>date_format</i>
<i>model_start_time</i>	optional	A time of day; default is <i>12:00:00</i>
<i>history_match_start_date</i>	optional	A date; default is <i>model_start_date</i>
<i>history_match_start_time</i>	optional	A time; default is <i>model_start_time</i>
<i>history_match_end_date</i>	optional	A date; default is end of simulation
<i>history_match_end_time</i>	optional	A time; default is end of simulation

Table 4.2 Keywords belonging to the GENERAL block of an OLPROC control file.

Keywords appearing in the GENERAL block of an OLPROC input file are now described in detail.

4.2 Date_format

The *date_format* keyword must be the first keyword provided in the GENERAL block of an OLPROC control file. It informs OLPROC of the date protocol used in its input files and output files.

The two options for *date_format* are as follows.

```
date_format = dd/mm/yyyy
```

and

```
date_format = mm/dd/yyyy
```

4.3 Time_for_CSV_files

Field measurements can be provided to OLPROC in either of two file types, namely CSV files saved from a spreadsheet program, or site sample files. The format for a site sample file requires that each line include a sample time to accompany the sample date. The same is not the case for measurement CSV files; for these files, times are optional.

As has already been discussed, when reading a measurement CSV file, OLPROC identifies the start of the second section of this file by the word “date” appearing in its second column. It then assumes that all columns starting from the second are grouped into pairs or triplets. The first column in each pair or triplet contains measurement dates. The last column contains measurement values. The optional middle column contains measurement times. If OLPROC finds the word “time” in the adjacent column to “date” (i.e. in the third column), then it expects triplets of measurement columns, with the second of each triplet devoted to time.

The *time_for_csv_files* keyword informs OLPROC what measurement time it should assume where measurements are recorded in doublets rather than in triplets of columns. This time can range between 00:00:00 and 23:59:59. If the *time_for_csv_files* keyword is omitted from the GENERAL block, OLPROC associates a time of 12:00:00 with all measurements.

4.4 Model_time_units

As is described in other sections of this document, OLPROC can read model-calculated quantities from two types of file. These are model-generated SSF files, and MODFLOW 6 OBS output files. The former type of file may have been written by a model postprocessor from the PEST Groundwater Utilities Suite.

As has already been described, site sample files associate a date and a time with each measurement. However, MODFLOW 6 OBS output files report simulation time only. OLPROC internally converts these simulation times to dates and time of day. In order to do this, it must know the simulation starting time and the time units employed by the model. OLPROC accepts two options for the latter, namely “days” and “seconds”. One of these text strings must follow the optional *model_time_units* keyword in the GENERAL block of an OLPROC control file. If this keyword is omitted, OLPROC assumes “days”.

4.5 Time_base_file

This keyword is optional, as is the provision of a time base file to OLPROC.

OLPROC can read model outputs from either of two types of file, namely SSF files and MODFLOW 6 OBS output files. In either case, model-generated values of system states and fluxes that are recorded in these files correspond to the times at which the model calculates them, normally the end of every model time step.

If a time base file is not supplied, then OLPROC insists that every model output file that it reads records model-generated values at identical times. Hence, collectively, these files contain a suite of model-generated time series that all employ the same time base. Each of these series has a name. This name is the name of the site that is associated with the series in an SSF file, or the observation name that is associated with the series in the header of a MODFLOW 6 OBS output file. OLPROC undertakes time-interpolation from the model time base to the times at which measurements were actually made; the latter times are supplied in the measurement CSV files and measurement SSF files that OLPROC reads.

A user can inform OLPROC that, when forming model-generated time series, it should employ only a subset of the times that are provided in model-generated SSF and MODFLOW 6 OBS output files. These times can be provided in a time base file. Such a file is exemplified in Figure 4.2.

```
# time base
0.0
100.0
200.0
300.0
400.0
500.0
etc
```

Figure 4.2. Part of a time base file.

As is apparent from Figure 4.2, the format of a time base file is simple. Like many other files read by OLPROC, it can include comment lines, these beginning with the “#” character. All other lines must contain a single time. Times must be listed in increasing order. They must correspond exactly to model output times. (As has already been discussed, the latter are expressed as simulation times in MODFLOW 6 OBS output files, and as dates and times in model-generated SSF files.) Times in a time base file must be expressed in the same units as those employed by the model. As discussed above, a user informs OLPROC of these units using the *model_time_units* keyword.

If you are unsure of model output times, check its output files. For MODFLOW, check its list output file. Alternatively, use programs from the PEST Groundwater Utilities Suite such as ARRDET, USGARRDET and MF6ARRDET to list times recorded in the headers to arrays that are stored in MODFLOW binary output files.

If a time base file is supplied, then it is not essential that times represented in all MODFLOW 6 OBS files and all model-generated SSF files be the same. However all of these model output files must provide data at all times that are recorded in the time base file. If a time base file is not supplied, then OLPROC will object if all model SSF and OBS output files which it is asked to read do not employ the same time base.

4.6 Model_start_date

A date must follow the *model_start_date* keyword. In conformity with the setting supplied with the *date_format* keyword, this date must be expressed using the protocol *dd/mm/yyyy* or *mm/dd/yyyy*. The *model_start_date* keyword is mandatory. It corresponds to simulation time zero.

4.7 Model_start_time

This optional keyword allows a user to nominate the time of day corresponding to simulation time zero. If it is not supplied, then 12:00:00 is assumed.

4.8 The History-Matching Window

Four optional keywords allow a user to define a time window for history-matching. Measurements supplied in measurement CSV and SSF files that are outside of this window are ignored by OLPROC. Hence model-generated quantities are not interpolated to these times. Nor do measurements that were made outside of this time window appear in the partial PEST control file that is written by OLPROC when run in “construction” mode.

Four keywords are used to define this window. They are as follows:

history_match_start_date

history_match_end_date

history_match_start_time

history_match_end_time

All of these keywords are optional. The following rules apply.

1. If a history-match start or end time is provided, the corresponding date must also be provided.
2. If a history-match start date is provided but a history-match start time is not provided, the latter is assumed to be 00:00:00.
3. If a history-match end date is provided but a history-match end time is not provided, the latter is assumed to be 24:00:00.
4. If a history-match start date is not provided, the history-match time window is assumed to begin at the start of the simulation.
5. If a history-match end date is not provided, the history-match time window is assumed to end at the final time of the model time base, this normally being the end of the simulation.

Note that the time spanned by the history-matching window can be greater than the model simulation time. This can be useful when part or all of the history-matching dataset corresponds to steady state conditions. Temporal interpolation to model output times outside of the simulation time window is discussed later in this manual.

5. MEAS_CSV_FILES Block

The MEAS_CSV_FILES block features only one keyword. However this keyword can be repeated many times. This keyword is *file*. The name of a measurement CSV file must follow a *file* keyword. This name can be surrounded by quotes. Quotes are mandatory if the filename contains a blank character. (The user is strongly encouraged NOT to include blank characters in filenames.)

See Figure 3.1 for an example of a MEAS_CSV_FILES block.

6. MEAS_SSF_FILES Block

6.1 General

The MEAS_SSF_FILES block is somewhat similar to the MEAS_CSV_FILES block, in that it provides a user with the means to nominate a set of files which OLPROC must read in order to obtain measurements for use in the history-matching process. However, because a site sample file (i.e. an SSF file) does not contain metadata, some of the information that is supplied in the leading section of a measurement CSV file must be supplied in the MEAS_SSF_FILES block of an OLPROC control file, along with the name of each SSF file.

As for measurements contained in a measurement CSV file, OLPROC allows both measurement absolutes and measurement differences to be included in the history-matching process. PEST observation names are formulated from measurement site names in the same way as for measurement CSV files - that is, through use of a “_N” or “_dN” suffix. As is explained in Section 2.3 of this document, the names of measurement sites are listed in the first column of an SSF file.

To simplify its control file protocol, OLPROC decrees that all absolute measurements contained in a single SSF file be assigned to the same observation group. Likewise, all measurement differences are assigned to the same observation group. The name of the former group is provided by the user in the OLPROC control file. The name of the latter group is obtained by affixing a “_d” suffix to the name of a group that is provided in the OLPROC control file.

6.2 The Protocol

The data entry protocol for a MEAS_SSF_FILES block is illustrated by Figure 6.1. This is extracted from Figure 3.1.

```
START MEAS_SSF_FILES
  file = gwh1.ssf  group=layer1 use_abs=yes use_diff=yes
  file = gwh2.ssf  group=layer2 use_abs=yes use_diff=yes
  file = gwf.ssf   group=flux   use_abs=yes use_diff=no
END MEAS_SSF_FILES
```

Figure 6.1 An example of the MEAS_SSF_FILES block of an OLPROC control file.

As is apparent from the above figure, four pieces of information must be provided on each line of the MEAS_SSF_FILES block. Optionally, any line can be continued by placing a “&” character at its end. This may enable easy reading of the information supplied on that line. The contents of the MEAS_SSF_FILE block shown in Figure 6.2 are the same as those shown in Figure 6.1.

```

START MEAS_SSF_FILES
  file = gwh1.ssf          &
    group=layer1          &
    use_abs=yes           &
    use_diff=yes
  file = gwh2.ssf          &
    group=layer2          &
    use_abs=yes           &
    use_diff=yes
  file = gwf.ssf           &
    group=flux            &
    use_abs=yes           &
    use_diff=no
END MEAS_SSF_FILES

```

Figure 6.2 The same MEAS_SSF_FILES block as in Figure 6.2, but with lines continued.

The name of an observation group must accompany the name of each measurement SSF file in a MEAS_SSF_FILES block of an OLPROC control file. The user must also indicate whether measurement absolutes and/or measurement differences will form part of the history-matching dataset. This is done by supplying “yes” or “no” with *use_abs* and *use_diff* keywords associated with the *file* keyword. Usage of absolutes and/or differences pertains to all measurements contained in the file. A suffix of “_d” is appended to the user-supplied observation group name in order to formulate the group name used by PEST for observation differences.

Beware of using differences in the calibration process in the unusual event that model-generated numbers are considerably separated in time from the first field measurement event. This may take you by surprise if, for some reason, model-generated numbers have NODATA values around this time.

7. MF6_OBS_FILES Block

7.1 General

OLPROC reads model-generated quantities from two types of file. These are:

- comma-delimited files produced by the MODFLOW 6 OBS package, and
- SSF files produced by model postprocessors.

As has already been discussed, these files must all provide model outputs at the same times, or at times that include those provided in a time base file.

Model outputs are matched to measurements through site (i.e. target) names. In measurement CSV files, these are provided in column headers; in measurement SSF files they are provided in its first column as site names.

A MODFLOW 6 OBS output file provides site/target names in its first row. For successful matching with site names provided in CSV and SSF measurement files, these names must be 20 characters or less in length. This requirement must be kept in mind when naming observations in a MODFLOW 6 input dataset.

A value of 3.0E30 in a MODFLOW 6 OBS output file has a special meaning; it signifies “no data”. On reading this type of file, OLPROC respects this convention. When undertaking temporal interpolation from model output times to measurement times, it accommodates “no data” values by ignoring them; it then adjusts temporal interpolation factors to account for their absence. If all values provided in a particular model-generated time series signify “no data”, OLPROC ceases execution with an appropriate error message.

Beware of using differences in the calibration process in the unusual event that model-generated numbers are considerably separated in time from the first field measurement event. This may take you by surprise if, for some reason, model-generated numbers have NODATA values around this time.

7.2 The Protocol

As is demonstrated in Figure 3.1, the protocol for supplying filenames in an MF6_OBS_FILE block is simple. The name of a file must follow a *file* keyword. Optionally an “=” symbol can separate the filename from the *file* keyword. If the filename contains a space, then it must be surrounded by double quotes.

8. MODEL_SSF_FILES Block

8.1 General

Model-generated SSF files respect the same protocol as measurement SSF files; see Figure 2.4. Programs of the PEST Groundwater Utilities Suite have the ability to read model output files of various types (including binary files), and record parts of their contents in site sample file format. In doing this, some of them undertake spatial interpolation from nodes of a model grid to the sites at which head, drawdown and concentration measurements were made. Others can accumulate boundary condition fluxes over many cells.

In similar fashion to its reading of a MODFLOW 6 OBS package output file, OLPROC interprets a value of 3.0E30 as “no data” (but only if the site sample file is read through the MODEL_SSF_FILES block). Interpolation to measurement times is adjusted accordingly.

8.2 The Protocol

The MODEL_SSF_FILES block of an OLPROC control file is illustrated in Figure 3.1. Each line of this block should provide a *file* keyword, an optional “=” symbol, and the name of a model-generated site sample file. If the latter contains a space (which is not recommended), its name should be surrounded by double quotes.

9. SECONDARY_MODEL_OUTCOMES Block

9.1 General

OLPROC supports calculation of “secondary model outcomes”. These can be matched with field measurements during the history-matching process.

Secondary model outcomes are calculated from “primary” model outcomes and other secondary model outcomes. Here we classify primary model outcomes as quantities that are featured in MODFLOW 6 OBS output files and in model SSF files. Each of these files records a time series of observations pertaining to one or more sites. A secondary model outcome associates a time series with a site whose name is specified by the user. Terms of this new time series are calculated from terms of existing time series using an equation. This same equation is applied to all terms comprising the series. The site with which the new series is associated can be new, and hence named by the user through the equation itself. Alternatively, the new series can be ascribed to an existing site; in the latter case, the existing time series associated with that site is over-written.

The time base associated with the new time series is the same as that associated with all model-generated time series. As has already been discussed, this can be read from a time base file, or is inherited from primary model-generated time series read by OLPROC.

Secondary model outcomes can serve a number of purposes. For example, the equation that assigns values to a secondary site may actually perform spatial interpolation from grid cell centres to the location of a measurement well. Alternatively, or as well, it may introduce a change of scale, and/or an average or summation of other series.

9.2 The Protocol

The SECONDARY_MODEL_OUTCOMES block that is illustrated in Figure 3.1 is repeated in Figure 9.1.

```
START SECONDARY_MODEL_OUTCOMES
  ff1= flux1 + flux2 + flux3
  hd1 = 0.5 * (bh3213 + bh3982)
  hd2 = min(head5,0.0)
  depth = 33.213 - (head1 + head2)/2.0
  flux5 = 2.3123
END SECONDARY_MODEL_OUTCOMES
```

Figure 9.1 Example of a SECONDARY_MODEL_OUTCOMES block of an OLPROC control file.

As is apparent from Figure 9.1, each line in a SECONDARY_MODEL_OUTCOMES block must contain an assignment equation. The equation can be of arbitrary complexity. Non-numeric variables featured on the right of an equation must be the names of sites that have been previously defined in MODFLOW 6 OBS output files, model SSF files, or in previous equations provided in the SECONDARY_MODEL_OUTCOMES block. As stated above, the equation is applied to all terms in the time series associated with these sites to generate corresponding terms in the new time series.

The term on the left of an equation can pertain to an existing site. Alternatively, it can define a new site. If it pertains to an existing site, then all terms of the time series associated with that site are overwritten. If it pertains to a new site, the usual naming convention limit of 20 characters must be observed when providing a name for the new site.

Here it is salient to remind the reader of a naming protocol to which previous reference was made in this document. Because any site associated with model-generated time series can potentially be used in an equation, the name of any site (measurement or model-generated) must not be such that it can be confused with a number.

If the name of a site contains a numerical operator then it must be surrounded by quotes if it is used in an equation. Otherwise quotes are arbitrary. Numerical operators are:

- “-“ (for subtraction);
- “+” (for addition);
- “*” (for multiplication);
- “/” (for division);
- “^” (for raising to a power); and
- “=” (for equality).

The equations shown in Figure 9.2 are legal or illegal as shown.

```
# legal equations
sitea = "site-a" + "site-b"
"sitea"="site-a"+"site-b"
"site-a" = "site-a" + "site-b"
# illegal equation
site-a=site-a + site-b
```

Figure 9.2 Example of legal and illegal equations.

9.3 Syntax for Equations

9.3.1 General

Much of the present subsection of this document is extracted from the manual for PLPROC. In doing so, it also repeats part of the manual for TS6PROC. The same equation functionality is provided by all of PLPROC, OLPROC and TS6PROC.

As discussed above, an equation provided in the SECONDARY_MODEL_OUTCOMES block of an OLPROC control file is applied repeatedly to corresponding elements of time series associated with all sites that are featured on the right side of the equation. The outcome of the equation provides a value for the corresponding element of the time series that is associated with the site that appears on the left side of the equation.

As will be discussed below, equations can also be provided in the WEIGHTS block of an OLPROC control file. In this case the equation may also be applied repeatedly; however an equation provided in the WEIGHTS block of an OLPROC control file is applied to observations rather than to elements of time series. Nevertheless the protocol employed for weights equations is the same as that described in the present section for time series equations.

9.3.2 Operators and Mathematical Functions

Arithmetic operators

Table 9.1 shows a list of arithmetic operators that can appear in an OLPROC equation. Note that the usual rules of operator precedence apply. If you are in doubt about precedence, use brackets to enforce your own order; inner bracketed expressions are always evaluated first.

Operator symbol	Operation
** or ^	<u>Power</u> . $a**b$ or a^b is interpreted as “a raised to the power b”.
/	<u>Division</u> . a/b is interpreted as “a divided by b”.
*	<u>Multiplication</u> . $a*b$ is interpreted as “a multiplied by b”.
-	<u>Subtraction</u> . This can be a unary or binary operator. $a-b$ is interpreted as “a minus b”; $-a$ is interpreted as “negative a”.
+	<u>Addition</u> . This can be a unary or binary operator. $a+b$ is interpreted as “a plus b”; $+a$ is interpreted as “positive a”.
()	<u>Brackets</u> . Terms within brackets are evaluated first. For example $5 + 4 * 3$ is evaluated as 17. However $(5 + 4) * 3$ is evaluated as 27.

Table 9.1. Arithmetic operators supported by OLPROC equations.

Logical operators

Logical operations produce an outcome that is either *TRUE* or *FALSE*. Logical operators supported by OLPROC are listed in Table 9.2.

Operator symbol	Operation
.lt. or <	<u>Less than</u> . $a.lt.b$ or $a < b$ is <i>TRUE</i> if a is less than b.
.le. or <=	<u>Less than or equal to</u> . $a.le.b$ or $a <= b$ is <i>TRUE</i> if a is less than or equal to b.
.eq. or ==	<u>Equal to</u> . $a.eq.b$ or $a == b$ is <i>TRUE</i> if a equals b.
.gt. or >	<u>Greater than</u> . $a.gt.b$ or $a > b$ is <i>TRUE</i> if a is greater than b.
.ge. or >=	<u>Greater than or equal to</u> . $a.ge.b$ or $a >= b$ is <i>TRUE</i> if a is greater than or equal to b.
.ne. or !=	<u>Not equal to</u> . $a.ne.b$ or $a != b$ is <i>TRUE</i> if a does not equal b.
.and. or &&	<u>And</u> . $a.and.b$ or $a \&\& b$ is <i>TRUE</i> if both a and b are true; for example $(1.lt.10).and.(6.lt.7)$ is <i>TRUE</i> .
.or. or	<u>Or</u> . $a.or.b$ or $a b$ is <i>TRUE</i> if a is <i>TRUE</i> or b is <i>TRUE</i> or both are <i>TRUE</i> ; for example $(1.lt.10).or.(1.gt.0)$ is <i>TRUE</i> .

!	<u>Not.</u> $!(a.lt.b)$ is <i>FALSE</i> if a is less than b .
---	---

Table 9.2. Logical operators supported by OLPROC.

Because the outcomes of OLPROC equations are assigned to elements of a site-associated time series, they must be real numbers. The outcome of an OLPROC equation with a logical outcome *TRUE* is converted to 1.0 before being assigned to a time series element; *FALSE* is converted to 0.0.

Mathematical functions

OLPROC equations support the following mathematical functions. The outcomes of all of these functions are real numbers.

Function	Definition
abs()	<u>Absolute value.</u> Argument can be any real number.
cos()	<u>Cosine.</u> Argument can be any real number supplied in radians.
acos()	<u>Inverse cosine.</u> Absolute value of argument must be less than or equal to one. Value is returned in radians.
sin()	<u>Sine.</u> Argument can be any real number supplied in radians.
asin()	<u>Inverse sine.</u> Absolute value of argument must be less than or equal to one. Value is returned in radians.
tan()	<u>Tan.</u> Argument can be any real number supplied in radians.
atan()	<u>Inverse tan.</u> Argument can be any real number. Value is returned in radians.
cosh()	<u>Hyperbolic cosine.</u> Argument can be any real number.
sinh()	<u>Hyperbolic sine.</u> Argument can be any real number.
tanh()	<u>Hyperbolic tan.</u> Argument can be any real number.
exp()	<u>Exponential.</u> Argument can be any real number.
log()	<u>Log to base e.</u> Argument must be a positive real number.
log10()	<u>Log to base 10.</u> Argument must be a positive real number.
sqrt()	<u>Square root.</u> Argument must be non-negative.
min(, ,)	<u>Minimum of a series of numbers.</u> Arguments can be any real numbers.
max(, ,)	<u>Maximum of a series of numbers.</u> Arguments can be any real numbers.
mod(,)	<u>Remainder.</u> $mod(a,b)$ is the remainder after a is divided by b .

Table 9.3. Mathematical functions supported by OLPROC equations.

If an equation requests an impossible operation (for example the square root of a negative number) OLPROC does not attempt to evaluate the equation; instead, it terminates execution with an error message that expresses its disgust at what it was asked to do.

10. What OLPROC Does

10.1 General

Now that many of the specifications of an OLPROC control file have been presented, it is time to provide a more detailed description of what OLPROC actually does with the information that it reads from this file. Specifications for the remainder of the OLPROC control file are provided in sections which follow the present one.

10.2 Measurement Sites and Observations

A user must provide OLPROC with time series of field measurements. These are recorded in one or a number of measurement CSV and SSF files. It is important to note that a measurement “time series” can be comprised of many measurements; alternatively, for one or many sites, it can comprised of a single observation. The latter facility allows OLPROC to be used in construction of PEST input datasets for steady state model calibration. This matter is further discussed below.

Where many measurements are provided for multiple sites, there is no need for these measurements to correspond to the same dates and times.

In creation of a PEST input dataset, OLPROC only processes elements of user-supplied measurement time series that fall within the history-matching window. This window can be explicitly defined using history-matching start and end dates/times supplied in the GENERAL block of an OLPROC control file. Alternatively, it can be implicitly defined by the starting date/time of the simulation and by the end date/time of the simulation. The latter is inferred from the last model output time associated with model-generated time series provided in MODFLOW 6 OBS output files and model output SSF files; alternatively it is the last time appearing in a user-supplied time base file. The beginning and end of the history-matching time window provided by a user can precede and/or postdate the beginning and end of the model simulation time window.

For any site (i.e. target) for which measurements exist within the history-matching window, OLPROC looks for model-generated data pertaining to the same site. Sites are linked by name; recall that names are case-insensitive. The names of model-generated sites are provided in MODFLOW 6 OBS output files, in model SSF output files and in the names of secondary time series defined in the SECONDARY_MODEL_OUTCOMES block of an OLPROC control file. If, for a particular measurement site, a corresponding model site cannot be found, OLPROC ceases execution with an appropriate error message.

10.3 New SSF Files

When run in “construction” mode, OLPROC writes a series of measurement SSF files. The contents of these files record measurements that were made within the history-matching window. However they omit measurements that are not specifically designated for use in the history-matching process through respective *use absolutes* and *use differences* headers in user-supplied measurement CSV files, and *use_abs* and *use_diff* keywords in the MEAS_SSF_FILES block of an OLPROC control file. For the “difference” case, differences are taken between each measurement and its respective reference measurement prior to

OLPROC recording this difference in the respective SSF file. The reference measurement for a particular site is the first measurement within the history-matching window for that site. Collectively, the contents of these OLPROC-generated SSF files therefore comprise observations which comprise the history-matching dataset. These files are referred to as “observation-generated SSF files” herein.

OLPROC writes one, or possibly two, such SSF files corresponding to each of the measurement CSV and measurement SSF files that are provided to it. The names of these files are formed from the names of the respective user-supplied measurement CSV and SSF files in the following manner:

- An *o_* prefix is added to each of these user-supplied filenames (“o” stands for observed).
- The dot which separates the filename base from the filename extension is replaced with an underscore.
- For files which hold time series of measurement differences, a *_d* suffix is appended to this filename string.
- A filename extension of *.ssf* is then provided.
- Optionally, these files can be stored in a user-nominated folder. If this is required, the name of this folder must be provided in the OUTPUT block of an OLPROC control file; see below.

When run in both “construction” and “postprocessor” modes, OLPROC writes another set of files which correspond to these observation-generated SSF files. These are referred to as “model-generated SSF files” herein. In fact, when run in “postprocessor” mode, production of these files is OLPROC’s only task.

The contents of model-generated SSF files are identical to those of observation-generated SSF files. That is, they feature the same sites and the same times. These times, of course, are times at which measurements were made. OLPROC interpolates from model output times to these measurement times in the following way.

- If a measurement precedes the first model output time, but is still within the history-matching window, then the value ascribed to its model-generated counterpart is the first model-generated value at that site. This corresponds to the first model output time.
- If a measurement postdates the last model output time, but is still within the history-matching window, then the value ascribed to the model-generated counterpart to this measurement is the last model-generated value at that site. This corresponds to the last model output time.
- For measurement times that fall between times at which model outputs are available, model outputs are linearly interpolated to those times.

The naming of model-generated SSF files follows an identical protocol to that described above for observation-generated SSF files. However their prefix is “*m_*” rather than “*o_*”.

10.4 Instruction Files

When OLPROC is run as a model postprocessor, the model-generated SSF files that it writes constitute “model outputs” from a PEST point of view. When run in “construction” mode, OLPROC writes instruction files through which PEST can read these model output files. These

instruction files have the same filename base as the files which they are designed to read. However the extension of *.ssf* is replaced by *.ins*.

10.5 Observation Names

In writing instruction files, OLPROC names each observation. As has already been stated, OLPROC generates observation names by appending the suffix “_N” or “_dN” to the site name for each element of the measurement time series associated with that site which lies within the history-matching window. The former suffix is employed for absolute observations while the latter suffix is employed for difference observations. *N* is the sequence number in the measurement time series. The number “1” corresponds to the first measurement within the history-matching window. Where differences are employed as observations, the number “1” is omitted, for this measurement is the reference observation which is subtracted from all other members of the time series to form differences.

In forming observation names, OLPROC respects the PEST name length convention of 20 characters or less. If necessary, it shortens the name of a site to accommodate the “_N” or “_dN” suffix corresponding to the highest *N* for that site. If this causes an observation name conflict, OLPROC reports the situation and ceases execution.

10.6 Observation Group Names

Observation group names are provided by the user on a site-by-site basis in measurement CSV files. They are provided on a file-by-file basis for measurement SSF files in the MEAS_SSF_FILE block of an OLPROC control file. If a user-supplied group name is greater than 10 characters in length, OLPROC objects. This accommodates the addition of a “_d” suffix for observation groups associated with measurement differences.

10.7 Partial PEST Control File

When run in “construction” mode, OLPROC writes a partial PEST control file. This file contains the following sections of a PEST control file:

observation groups

observation data

model command line

model input/output (part thereof)

It is the user’s job to add the other sections of the PEST control file to these OLPROC-written sections, or to blend the contents of an OLPROC-produced partial PEST control file with those of an existing PEST control file. When doing this, careful attention must be paid to assigning correct values to the NOBSGP, NOBS and NINSFLE variables in the “control data” section of the new PEST control file. As is explained in PEST documentation, these record the number of observation groups, number of observations and number of instruction files featured in a PEST control file. In order to assist the user with this task, OLPROC records the values of these variables on the screen as it ceases running in “construction” mode.

10.8 Additions to the Model Batch File

In writing the “model command line” section of a PEST control file, OLPROC assumes that the model is run using a batch or script file whose name is provided in the OUTPUT block of the OLPROC control file. If this name is not provided, OLPROC uses the name *model.bat*. The name of this batch or script file is supplied as the command to run the model. Presumably, this batch or script file will contain commands to run not just the simulator, but one or more preprocessors such as PLPROC and TS6PROC to undertake various tasks required for model parameterization.

It is the responsibility of you, the user, to add the command to run OLPROC to the model batch file. This command should be placed after that to run the model. The command is comprised of the following text:

```
OLPROC filename 1
```

where *filename* is the name of the same OLPROC control file that was used to run OLPROC in “construction” mode. As was described in Section 1 of this manual, the “1” instructs OLPROC to run in “postprocessor” mode. When run in this mode, its only task is to read MODFLOW 6 OBS output CSV files, and/or model SSF files, and to write its own set of model-generated SSF files in the manner described above.

It is important to note that, even when run in “postprocessor” mode, OLPROC still needs to read all of the measurement files that are named in its control file. Hence these must be accessible to OLPROC when it is run in this mode. For convenience, these files may be placed in the model working folder, along with files that are read and written by the model itself. When PEST or PEST++ undertakes model runs in parallel, they must also be transferred to agent folders so that these agents can also run the OLPROC postprocessor through the model batch file.

10.9 Steady State History-Matching

OLPROC can be used to facilitate history-matching of both steady state and transient models. As stated above, where steady state history-matching is undertaken, site-associated “time series” (either measured or model-generated) contains only one element. OLPROC has no trouble in accommodating this situation. However it is important to ensure that the time associated with each user-supplied measurement lies within the history-matching time window that is provided to OLPROC; this measurement is thus included in the history-matching dataset. As is apparent from the above description of OLPROC’s time interpolation algorithm, model-generated counterparts to steady state field measurements can readily be obtained from model results pertaining to a single output time.

11. OUTPUT Block

11.1 General

As presently programmed, only three keywords can be supplied in the OUTPUT block of an OLPROC control file; only one of these is mandatory. These keywords are now described.

The OUTPUT block that appears in the OLPROC control file that is illustrated in Figure 3.1 is repeated in Figure 11.1.

```
START OUTPUT
  partial_pest_control_file = partial.pst
  model_batch_file = runmodel.bat
  obs_ssf_folder = .\obs_files
END OUTPUT
```

Figure 11.1 An OUTPUT block of an OLPROC control file.

11.2 Partial PEST Control File

The name of the partial PEST control file that OLPROC must write when run in “construction” mode must be provided following the mandatory *partial_pest_control_file* keyword. If this keyword is omitted, OLPROC will cease execution with an error message.

11.3 Model Batch File

Supply the command to run the model following the *model_batch_file* keyword. This must be the name of a batch or script file. As discussed in the previous section of this manual, it is the user’s responsibility to add the command to run OLPROC in “postprocessor” mode to this batch or script file, following the command to run the model.

If the *model_batch_file* keyword is omitted, OLPROC uses “model.bat”.

11.4 Folder for Observation-Generated SSF Files

The *obs_ssf_folder* keyword is optional. If it is omitted, then observation-generated SSF files written by OLPROC when it runs in “construction” mode are recorded in the folder from which OLPROC is run. On most occasions this will be the model working folder.

As already discussed, observation-generated SSF files are not actually used in the inversion process. They are written for the user’s convenience. They comprise the observed counterparts to model-generated SSF files written by OLPROC when it runs in both “postprocessor” and “construction” modes. If a user wishes that these files be stored in a different folder from the OLPROC working folder, the name of that folder can be provided following the *obs_ssf_folder* keyword. Note that relative pathnames are permitted. Hence the name of the folder can begin with a dot.

12. Weights Block

12.1 General

As is described in PEST documentation, the *observation data* section of a PEST control file lists measured values to which model outputs must be fit through parameter adjustment. It also lists weights that are used in formulation of the objective function that PEST employs to denote model-to-measurement misfit.

The WEIGHTS block of an OLPROC control file contains a series of directives through which a user informs OLPROC how to calculate and assign weights to observations. This block is optional. If it is not provided, then a value of 1.0 is ascribed to all observation weights that pertain to measurement absolutes, while a value of 100.0 is ascribed to all observation weights that pertain to measurement differences. These same default values are ascribed to observations and observation groups for which all directives in the WEIGHTS block are inapplicable.

There is no limit on the number of directives that can appear in the WEIGHTS section of an OLPROC control file. Each directive supersedes those which precede it. When OLPROC constructs a partial PEST control file, it reads the directives contained in the WEIGHTS block in reverse order on each occasion that it records an observation and corresponding weight in this file. As soon as it finds a directive that pertains to that particular observation, it implements that directive; it then moves on to the next observation.

For reference, the WEIGHTS block pertaining to the OLPROC control file illustrated in Figure 3.1 is repeated below.

```
START WEIGHTS
  name          = drawdown                &
    date1       = 1/1/1990 date2=1/1/2000  &
    weight      = 1.0/(obsval+0.005)        &
    minweight   = 0.001                    &
    maxweight   = 10.0                     &
    type        = diff
  name = heads type=diff weight=5.0
END WEIGHTS
```

Figure 12.1. An example of a WEIGHTS block of an OLPROC control file.

Any line within a WEIGHTS block can be continued to the next line by placement of a “&” character at the end of the line. This is illustrated in Figure 12.1. The same information that is conveyed in Figure 12.1 is conveyed without the use of continuation characters in Figure 12.2.

```
START WEIGHTS
  name=drawdown date1=1/1/1990 date2=1/1/2000 weight=1.0/(obsval+0.005) minweight = 0.001 maxweight = 10.0 type=abs
  name = heads type = diff weight=5.0
END WEIGHTS
```

Figure 12.2. The same WEIGHTS block as in Figure 12.2, but without line continuation.

12.2 Protocol

12.2.1 Keywords

Up to eight keywords can be supplied on each line of a WEIGHTS block. Only three are mandatory, these being *name*, *type* and *weight*; the remainder are optional. Each keyword must be followed by an “=” symbol. The keywords are as follows:

name
type
weight
date1
time1
date2
time2
minweight
maxweight

12.2.2 Name

Each line within a WEIGHTS block must begin with the string “*name*=”. (A space can precede and follow the “=” symbol if desired.) The name of a measurement site or observation group must follow this string. (Note that, in order to avoid confusion in assigning weights to observations or observation groups, OLPROC will object with an error message if the user supplies an observation group name which is the same as the name of a site.) If a site name follows the *name* keyword, this name must have been provided to OLPROC through a measurement CSV file or a measurement SSF file. The name of an observation group must have been provided through a measurement CSV file, or following a *group_name* keyword in the MEAS_SSF_FILE block of an OLPROC control file.

If the name of an observation group is supplied, then the weights strategy provided on a particular line of the WEIGHTS block is applied to all members of that group. Note that the group name must not include the “_d” suffix that OLPROC automatically affixes to groups comprised of measurement differences. Instead, the *type* keyword is used to distinguish between differences and absolutes.

12.2.3 Type

The value ascribed to *type* must be either “abs” or “diff”. This restricts application of the weighting strategy supplied in a weights directive to either absolute measurements or differences of measurements.

12.2.4 Weights

Text following the “*weight*=” string can comprise a number or an equation. If it is a number, then this number must not be less than zero. Alternatively, an equation can be of arbitrary complexity (see section 9.3); however it can cite only one variable, this being *obsval*. “*Obsval*” stands for “observation value”. Thus weights can be formulated as functions of measured values, or as functions of measured differences, this depending on whether *type* is characterized as “abs” or “diff”.

Care must be taken in formulating a weights equation. In particular:

- Ensure that the equation is calculable regardless of the value of a measurement or measurement difference. Safeguards can be put in place to protect against values of zero being assigned to equation denominators or to functions such as *log()* which will not accept this value. Thus the equation $1/(obsval+0.001)$ is more robust than the equation $1/obsval$.
- Ensure that the outcome of a weights equation is never negative.

12.2.5 Maximum and Minimum Weights

The *maxweight* and *minweight* keywords are optional. If evaluation of a weights equation results in a weight whose value is greater than *maxweight*, then the weight is clipped at *maxweight*. If it results in a weight whose value is less than *minweight*, then the weight is clipped at *minweight*. If *minweight* is not supplied, it is awarded the default value of 0.0.

12.2.6 Time Interval

Assignment of a weight, or weights equation, to a particular measurement site or group of sites can be restricted to a particular time interval. This is done using the *date1*, *time1*, *date2* and *time2* keywords. All of these are optional. The following rules apply.

- If a *time1* keyword is supplied, then a *date1* keyword must also be supplied.
- If a *time2* keyword is supplied, then a *date2* keyword must also be supplied.
- If the *date1* keyword is omitted, it is assumed to coincide with the start of the history-matching window.
- If the *date2* keyword is omitted, it is assumed to coincide with the end of the history-matching window.
- If a *time1* or *time2* keyword is omitted, then the time associated with *date1* is assumed to be 00:00:00, while the time associated with *date2* is assumed to be 24:00:00.
- *date2/time2* must postdate *date1/time1*.